Compulsory Assignment 2

Group 97 - Predictive Analytics (KAN-CDSCO1005U)

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Part 1

1.1 Time Series Plot, ACF, and Decomposition

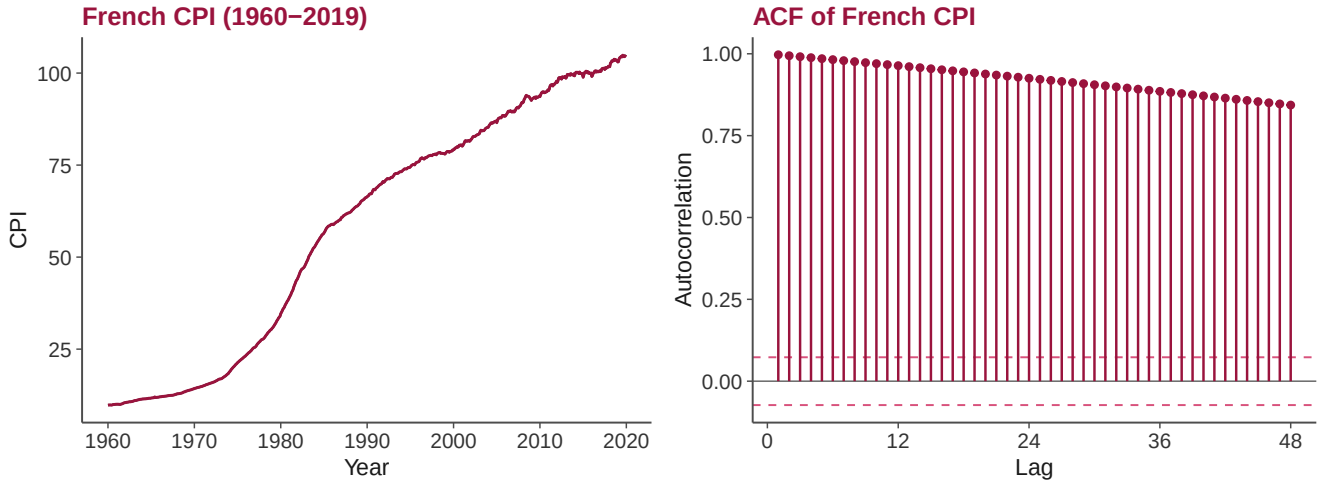

Figure 1: Left: French CPI from January 1960 to December 2019. Right: sample ACF up to 48 lags.

The CPI series displays a strong upward trend with no mean reversion. Growth is steeper in the 1970s–80s (oil crises) and flattens from the mid-1980s onward. The variance increases with the level, suggesting a variance-stabilising transformation is needed. The ACF decays extremely slowly, confirming the series is non-stationary and must be differenced before ARIMA modelling.

We apply a classical multiplicative decomposition because the seasonal amplitude grows with the level of the series. This method provides a straightforward separation of trend, seasonal, and irregular components as a first exploratory step.

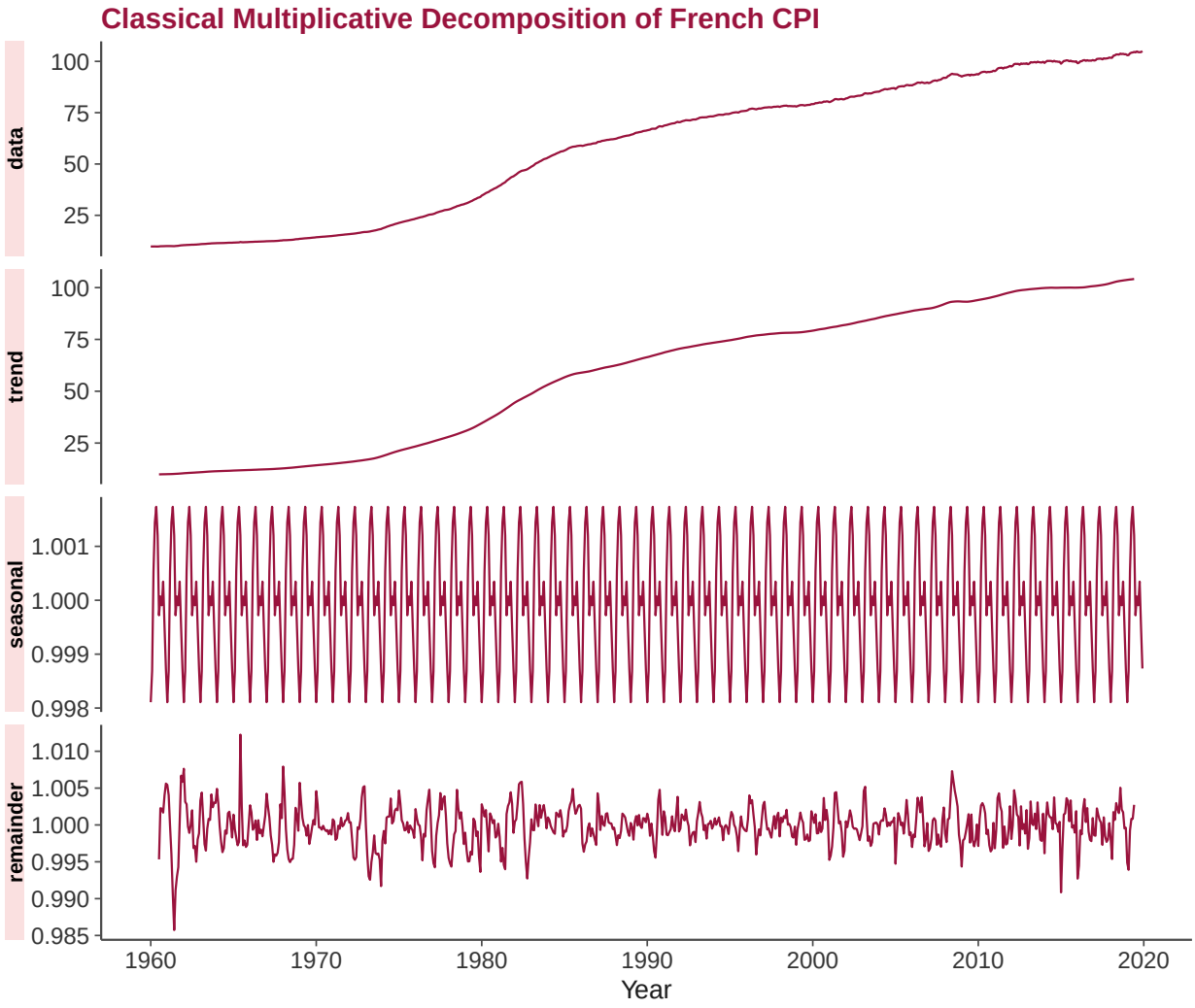

Figure 2: Multiplicative decomposition into data, trend, seasonal, and remainder.

The decomposition reveals a smooth upward trend, a regular annual seasonal pattern, and a remainder that is largest during the 1970s inflation. The method assumes a constant seasonal pattern and uses symmetric moving averages (losing the first/last six months), which are its main limitations over a 60-year span with structural change.

1.2 Transformation

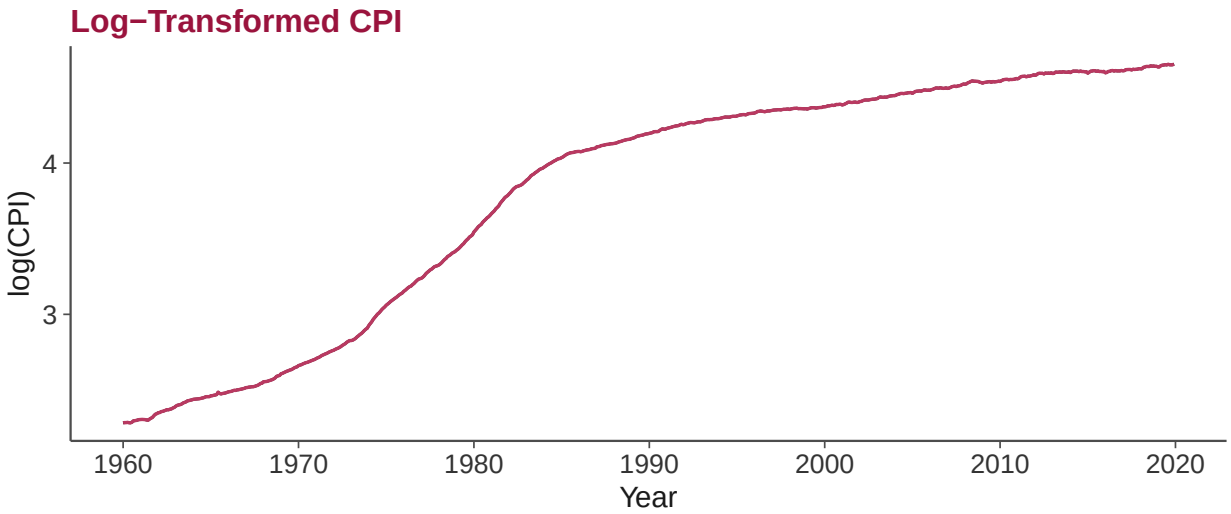

Figure 3: Log-transformed French CPI series.

The Box-Cox method yields $\hat{\lambda} = 1.32$, close to zero, confirming that the natural logarithm is near-optimal for variance stabilisation. After the log transformation, variance is stable and the seasonal fluctuations become approximately constant in magnitude, making an additive decomposition appropriate. We use the **log-transformed series for the remainder of the assignment** because $\lambda \approx 0$ and first differences of log(CPI) approximate percentage changes, aiding economic interpretation.

1.3 Unit Root Tests (KPSS and ADF)

The log-transformed CPI has a clear deterministic trend, so we use KPSS with `type = "Trend"` and ADF with `k = 12` (monthly lag structure) to test for stationarity.

Table 1: Unit root tests on log(CPI).

Test	Test Statistic	p-value
KPSS (Trend)	2.4629	0.01
ADF	-1.8044	0.661

Table 1: Unit root tests on log(CPI).

KPSS rejects trend-stationarity ($p < 0.01$) and ADF fails to reject the unit root null ($p = 0.661$). Both tests consistently indicate log(CPI) contains a unit root.

1.4 Making the Series Stationary and ARIMA Identification

We apply first-order regular differencing ($d = 1$) and first-order seasonal differencing ($D = 1$, lag 12) to remove the stochastic trend and seasonality.

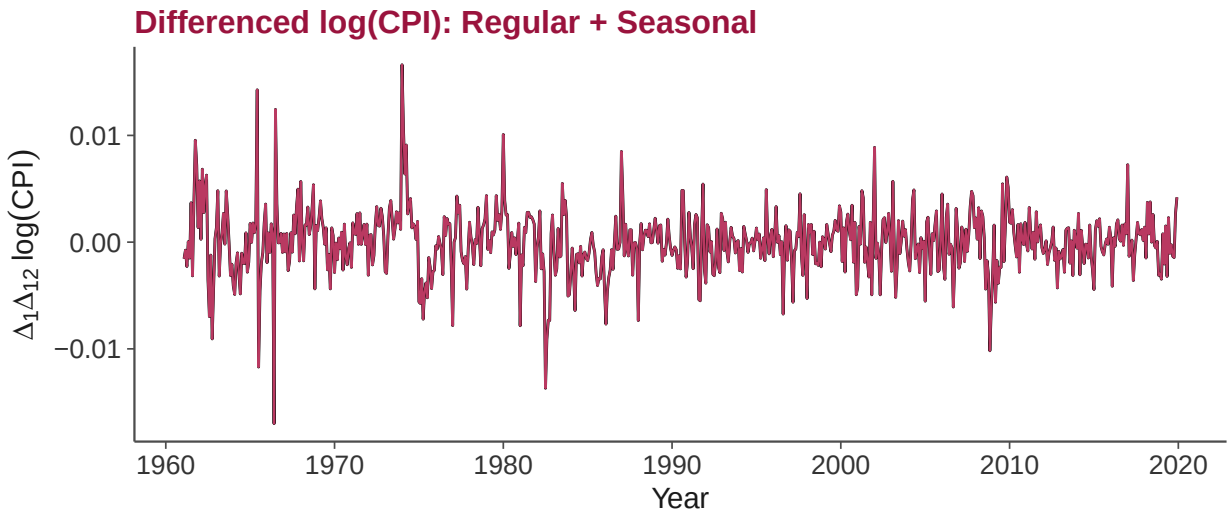

Figure 4: First regular and first seasonal difference of log(CPI).

The KPSS test on the differenced series gives a statistic of 0.1268 ($p = 0.1$), so we do not reject level stationarity.

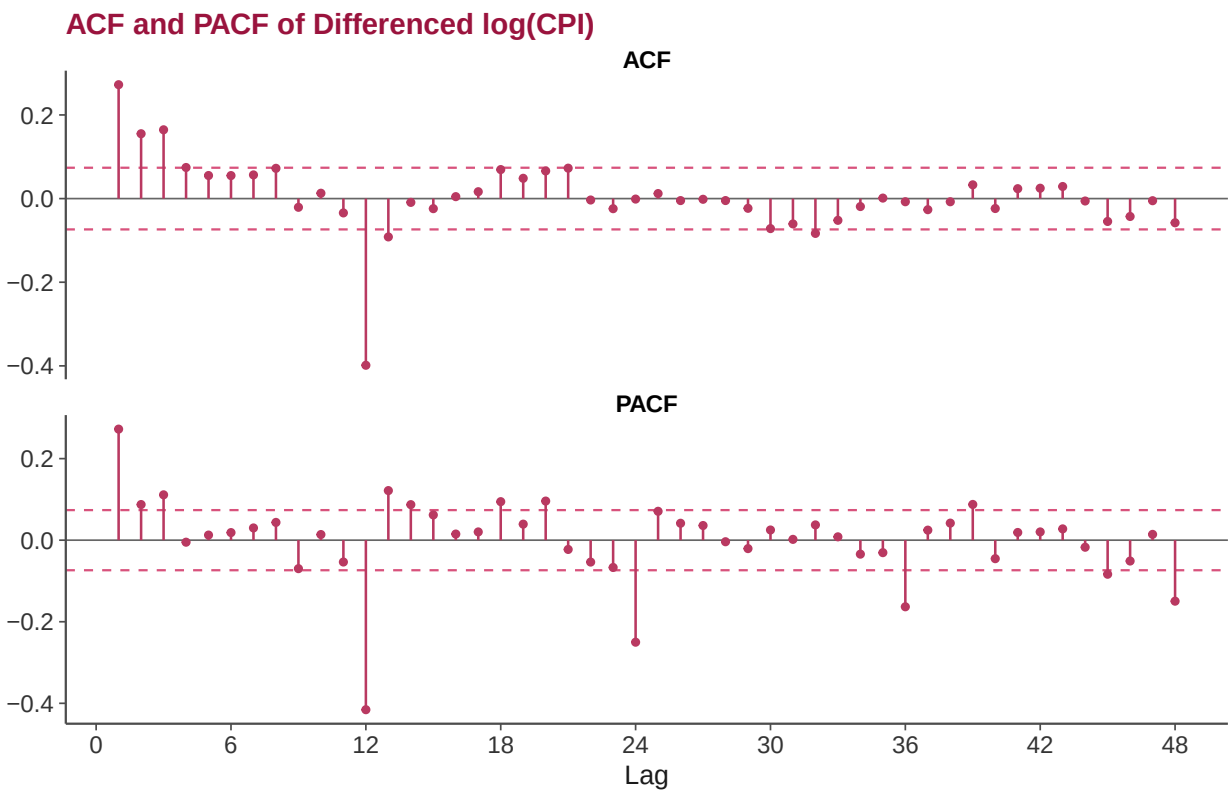

Figure 5: ACF (top) and PACF (bottom) of the doubly differenced log(CPI).

The ACF shows a significant spike at lag 1 that cuts off, suggesting MA(1). At seasonal lag 12 it shows a spike that decays, suggesting seasonal MA(1). The PACF is consistent with this MA interpretation. We propose the "airline model": ARIMA$(0,1,1)(0,1,1)_{12}$, written as $(1-B)(1-B^{12})\log Y_t = (1+\theta_1 B)(1+\Theta_1 B^{12})\varepsilon_t$.

Part 2

2.1 Train/Test Split and Auto ARIMA

We reserve the last 24 months (Jan 2018–Dec 2019) as the test set (24 obs), leaving 696 observations for training.

Using `auto.arima` with exhaustive AIC search, the selected model is **ARIMA(1,2,2)(2,0,0)[12]**. This differs from our manual specification, suggesting that additional AR and seasonal AR terms improve in-sample AIC.

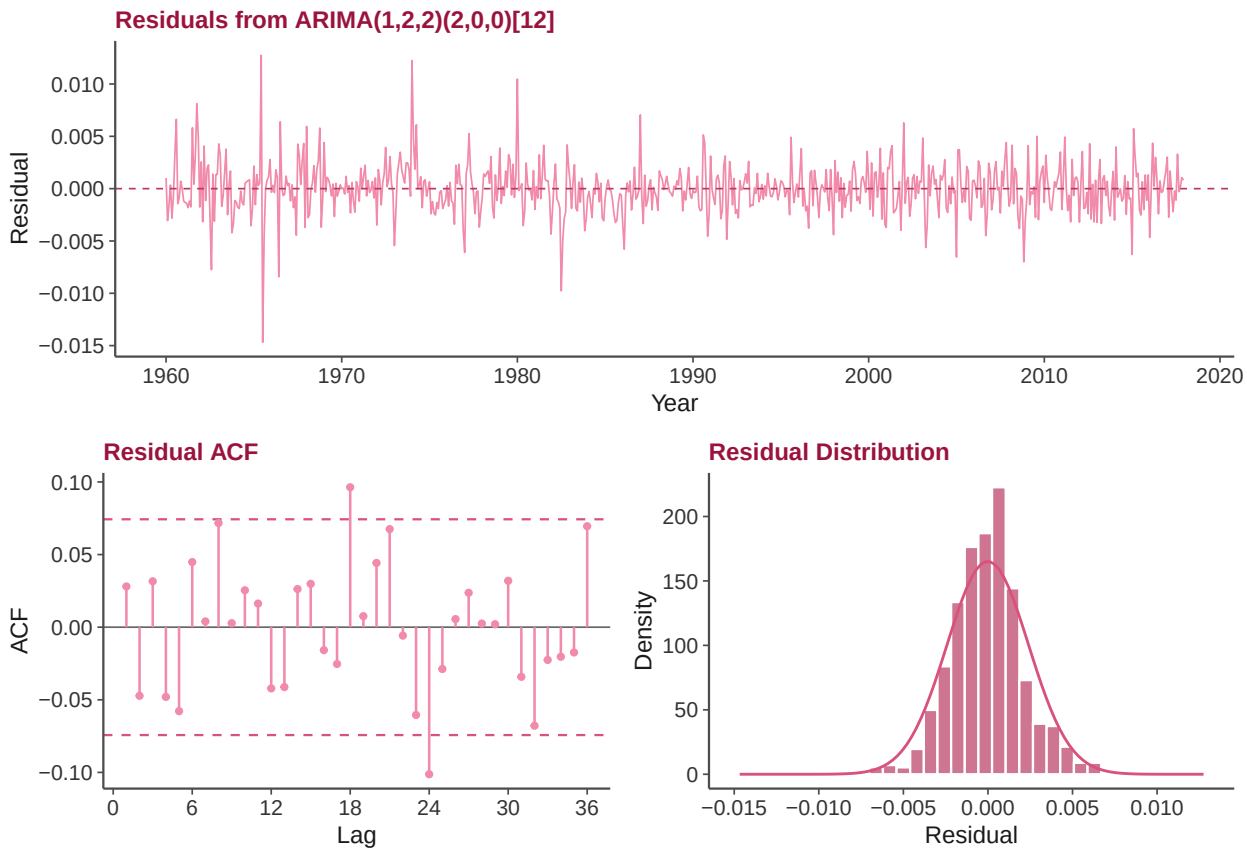

Figure 6: Residual diagnostics. Top: time series. Bottom-left: ACF. Bottom-right: histogram with normal overlay.

The Ljung-Box test at 24 lags gives $Q^* = 38.18$ ($p = 0.00563$), indicating some residual autocorrelation at the 5% level. The largest deviations concentrate in the high-inflation 1960s–70s. The histogram is approximately normal but slightly heavy-tailed. The model could be improved with intervention dummies for structural breaks, but the overall residual magnitude is small and the fit is adequate.

2.2 Forecasting and Model Comparison

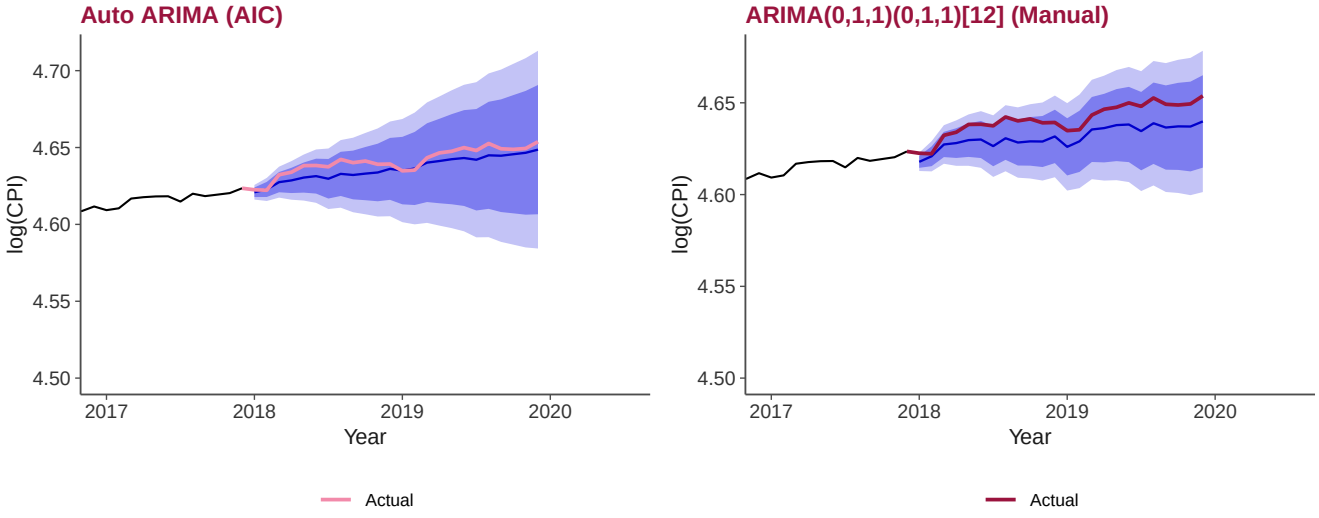

Figure 7: 24-month out-of-sample forecasts (zoomed, 2015-2020). Shaded bands show 80% and 95% prediction intervals.

Table 2: Forecast accuracy comparison.

Model	Set	RMSE	MAE	MAPE	MASE
Auto ARIMA (AIC)	Training set	0.0024	0.0017	0.0499	0.0426
Auto ARIMA (AIC)	Test set	0.0056	0.0050	0.1075	0.1220
ARIMA(0,1,1)(0,1,1)[12]	Training set	0.0025	0.0018	0.0522	0.0441
ARIMA(0,1,1)(0,1,1)[12]	Test set	0.0101	0.0096	0.2071	0.2350

Table 2: Forecast accuracy — Auto ARIMA vs. Manual ARIMA(0,1,1)(0,1,1)[12].

The Auto ARIMA model achieves a lower test-set RMSE and MAE, indicating that the additional parameters translate into better 24-month forecasting performance. Both models have a MASE well below 1, comfortably outperforming a naïve seasonal benchmark. The Auto ARIMA is the preferred model for forecasting accuracy, though the simpler airline model offers greater parsimony.